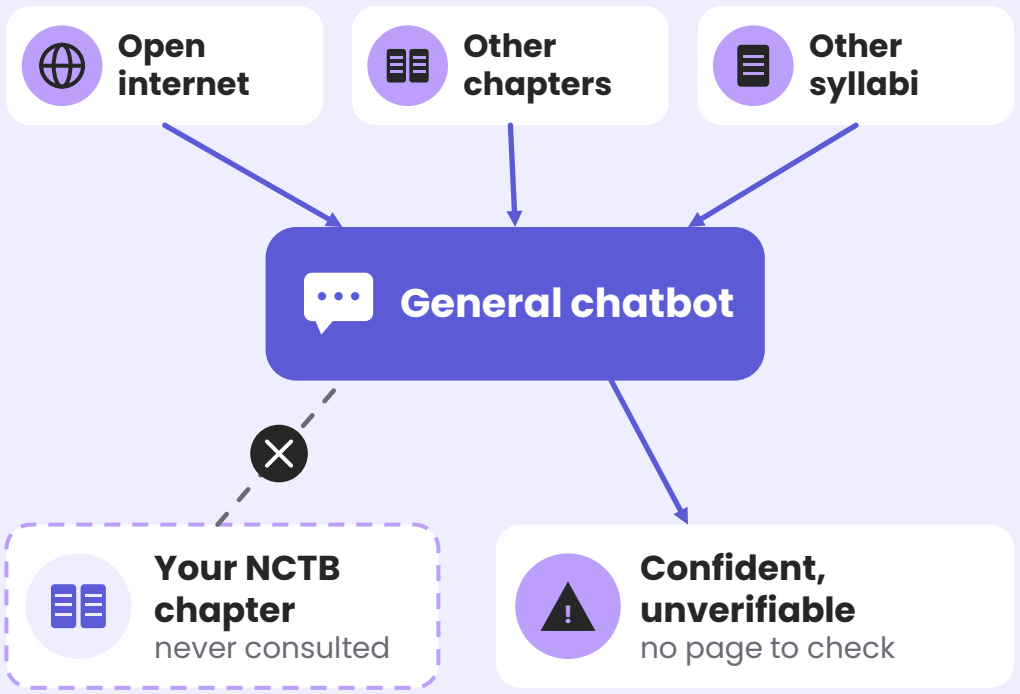
PROBLEM STATEMENT

Generic AI fails exam prep

Where a general chatbot's answer comes from



AI makes things up

Answers come from the open web and drift off-syllabus.

No self-check

Students can't test one specific chapter.

No chapter quizzes

Teachers can't get a fresh quiz per chapter on demand.

Parents in the dark

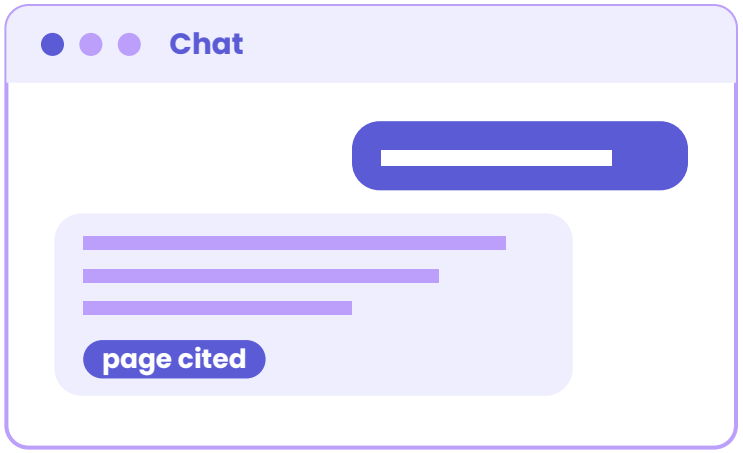
Nobody sees which chapter or topic a child is weak in.

PROPOSED SOLUTION

One chapter, three study tools

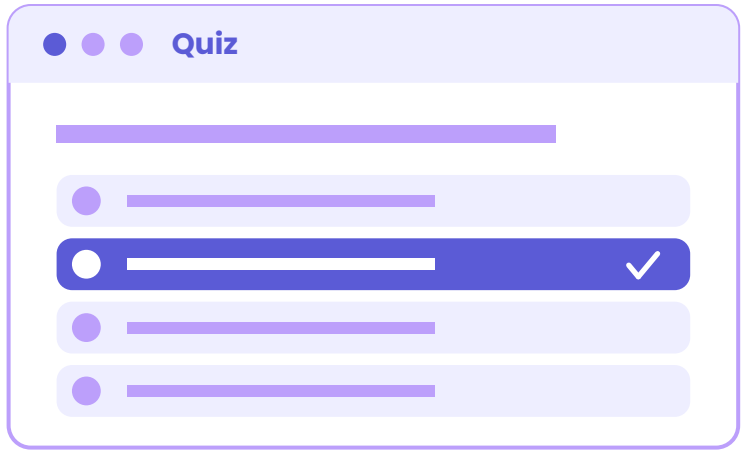
Class 9–10 → Pick a subject → Pick a chapter

Everything below stays inside that one chapter.



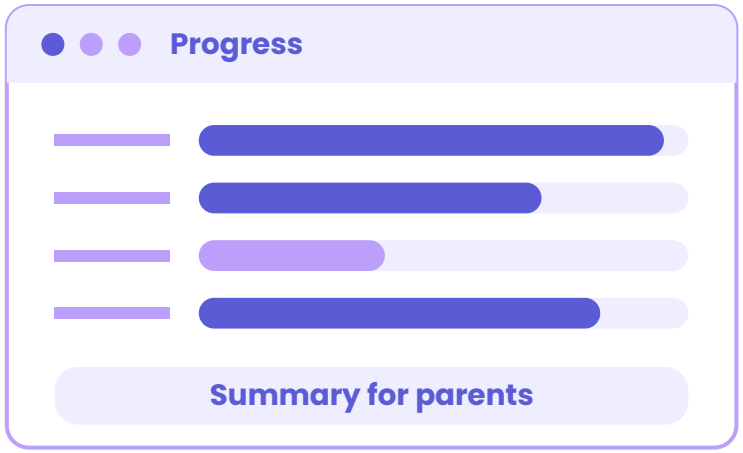
Chat

Ask in English or বাংলা. Streamed answers cite the page.



Quiz

5 multiple-choice questions with explanations and a score.



Progress

Chapter accuracy, weakest topics, and a summary for parents.

EN ⇄ বাং **Bilingual by design: one toggle switches the interface, subjects, chapter titles and AI answers.**

HOW THE SOLUTION WORKS

From textbook PDF to cited answer

Built For Ostad presents YUNet Algnite BUILDATHON 2026 National AI Hackathon for B2C Solutions

OFFLINE · BUILT ONCE

NCTB PDFs
27 textbooks as PDF pages

→

OCR
Text extracted from every page

→

Chunking
1500 characters, 350 overlap

→

Embeddings
multilingual-e5-base, 768-d

→

FAISS index
One per chapter + chunks.json

search runs inside the chosen chapter; nothing is re-embedded

LIVE · EVERY QUESTION

Question
English or বাংলা, in one chapter

→

Hybrid search
BM25 0.4 + FAISS 0.6, k=5

→

Generation
Groq gpt-oss-120b, DeepSeek fallback

→

Streaming
NDJSON events, token by token

→

Cited answer
Page reference shown to student

Quiz can't be cheated
Correct answers never leave the server.

Progress is stored
SQLite (WAL): one row per answered question.

Docker on AWS EC2
Backend: AWS
Frontend: Vercel

Built For Ostad presents YUNet Algnite BUILDATHON 2026 National AI Hackathon for B2C Solutions

INTRODUCTION · TEAM BACKGROUND

Meet the solo builder



Md. Tanvir Sheikh

AI/ML engineer · Dhaka, Bangladesh

- tanvirsheikh0177@gmail.com
- github.com/Tanvir-Sheikh-R
- linkedin.com/in/tanvir-sheikh-r
- Solo participant

Background

CSE degree
Final semester exams completed

ML/AI internship
FlyRank AI

AI/ML self-study
About 1.5 years, Apna College Prime course

Video editing
Prior work experience at Udvash

What I built, end to end

- **Data pipeline**
- **Backend API**
- **Frontend**
- **Deployment**

OTHER KEY INFORMATION

Key facts at a glance

27
books

385
searchable chapters

437
titles translated to English

5
questions per quiz

Privacy and security

- No accounts, no sign-up, no personal data.
- Anonymous browser ID; chats deleted after 3 days.
- Quiz answers stay on the server.

Quality and speed

- Model cache 1.1 GB.
- Catalog served with ETag, gzip and browser cache.
- End-to-end tests pass in English and Bangla.

What's next

- More classes beyond Class 9–10.
- Teacher dashboard.
- Offline PWA.

Live app: boibondhu.vercel.app API health: http://13.48.12.56:8000/api/health